

# tmux // Terminal Multiplexer

Complete Technical Architecture, Control Algebra, Vi Buffers & Hardened Configuration

## 01. ARCHITECTURAL TOPOLOGY & PTY VIRTUALIZATION

### Client-Server Execution Decoupling

tmux runs as an asynchronous event-driven daemon ('tmuxd'), detaching child processes from the originating GUI terminal or remote SSH session.



## 02. CONTROL MATRIX & KEYBINDING ALGEBRA

### Session Lifecycle

GLOBAL

| ACTION           | CLI COMMAND                                                 | INTERACTIVE                                       |
|------------------|-------------------------------------------------------------|---------------------------------------------------|
| New Session      | <code>tmux new -s &lt;name&gt;</code>                       | <code>:new -s</code><br><code>&lt;name&gt;</code> |
| Attach Target    | <code>tmux attach -t</code><br><code>&lt;name&gt;</code>    | —                                                 |
| Detach Client    | <code>tmux detach</code>                                    | Prefix <code>d</code>                             |
| Interactive Tree | <code>tmux choose-tree -s</code>                            | Prefix <code>s</code>                             |
| Rename Session   | <code>tmux rename-session</code><br><code>&lt;n&gt;</code>  | Prefix <code>\$</code>                            |
| Kill Session     | <code>tmux kill-session -t</code><br><code>&lt;n&gt;</code> | <code>:kill-session</code>                        |

### Window Virtualization

TABS

| ACTION               | CLI / COMMAND                                                | INTERACTIVE                            |
|----------------------|--------------------------------------------------------------|----------------------------------------|
| Create Window        | <code>tmux new-window -n</code><br><code>&lt;n&gt;</code>    | Prefix <code>c</code>                  |
| Next / Previous      | <code>tmux next-window /</code><br><code>prev</code>         | Prefix <code>n</code> / <code>p</code> |
| Direct Slot Jump     | <code>tmux select-window -t</code><br><code>&lt;i&gt;</code> | Prefix <code>0..9</code>               |
| Interactive Selector | <code>tmux choose-tree -w</code>                             | Prefix <code>w</code>                  |
| Rename Window        | <code>tmux rename-window</code><br><code>&lt;n&gt;</code>    | Prefix <code>,</code>                  |
| Terminate Window     | <code>tmux kill-window</code>                                | Prefix <code>&amp;</code>              |

Pane Geometry & Geometry Tiling

ERGONOMIC

| OPERATION                     | DEFAULT BINDING                      | OPTIMIZED<br>(.TMUX.CONF) | CORE SYSTEM COMMAND                                     |
|-------------------------------|--------------------------------------|---------------------------|---------------------------------------------------------|
| Vertical Split (Side-by-Side) | Prefix %                             | Prefix                    | <code>split-window -h -c "#{pane_current_path}"</code>  |
| Horizontal Split (Stacked)    | Prefix "                             | Prefix -                  | <code>split-window -v -c "#{pane_current_path}"</code>  |
| Directional Focus Navigation  | Prefix Arrows                        | Prefix h j k<br>1         | <code>select-pane -[U D L R]</code>                     |
| Zoom / Fullscreen Toggle      | Prefix z                             | Prefix z                  | <code>resize-pane -Z</code> (preserves layout state)    |
| Cycle Tiling Layouts          | Prefix Space                         | Prefix Space              | <code>next-layout</code> (even-horiz, even-vert, tiled) |
| Synchronize Panes (Broadcast) | <code>:setw synchronize-panes</code> | Prefix y                  | Sends simultaneous keystrokes across all panes          |

03. THE VI BUFFER ENGINE & CLIPBOARD PIPELINE

Vi Copy Mode Pipeline

MEMORY BUFFER

Scrollback is stored in server-allocated memory, isolated from terminal viewport limitations.

1. Enter Mode:

Prefix

[

2. Vi Navigation:

h

j

k

l

,

Ctrl+u

,

Ctrl+d

3. Search:

/

(forward)

,

?

(backward)

4. Visual Select:

v

5. Yank (Copy):

y

→ passes to OS clipboard & buffer

6. Paste Buffer:

Prefix

]

Failure Remediation

DIAGNOSTICS

Vim Esc Delay

Set

escape-time 0

in config to remove modal delay.

Color Loss

Enforce

default-terminal "tmux-256color"

+ RGB overrides.

Nested SSH

Press

Prefix

twice to pass hotkey to guest session.

Small Viewport

Enable

setw -g aggressive-resize on

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04. HARDENED PRODUCTION CONFIGURATION (~/.TMUX.CONF)

```
# =====
# PhTex High-Performance Production .tmux.conf (Light Theme Build)
# =====

unbind C-b
set -g prefix C-a                                # Ergonomic prefix mapping
bind C-a send-prefix

set -g default-terminal "tmux-256color"          # Full 24-bit TrueColor RGB
set -ag terminal-overrides ",xterm-256color:RGB"
set -s escape-time 0                             # 0ms delay for Neovim/Vim
set -g history-limit 50000                       # 50,000 Line scrollback buffer
set -g mouse on                                  # Full mouse support

set -g base-index 1                             # 1-based indexing for windows & panes
setw -g pane-base-index 1
set -g renumber-windows on

# Intuitive Split & Directory Preservation
bind | split-window -h -c "#{pane_current_path}"
bind - split-window -v -c "#{pane_current_path}"

# Vi Navigation Chords
bind h select-pane -L \; bind j select-pane -D \; bind k select-pane -U \; bind l select-pane -R
setw -g mode-keys vi
bind -T copy-mode-vi v send-keys -X begin-selection
bind -T copy-mode-vi y send-keys -X copy-pipe-and-cancel "wl-copy || xclip -in -selection clipboard"

# Minimalist Status Bar Design
set -g status-style "bg=#e2e8f0,fg=#0f172a"
set -g status-left "[bg=#4f46e5,fg=ffffff,bold] S [bg=#e2e8f0,fg=#4f46e5] "
set -g window-status-current-format "[fg=ffffff,bg=#059669,bold] #I:#W "
bind r source-file ~/.tmux.conf \; display-message "Config Reloaded."
```